



Low-Light Image Enhancement via Multi-Scale Feature Fusion

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ENTER



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01.

INTRODUCTION



What is this Project ?



PROJECT INTRODUCTION



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- A locally hosted web application built with multi-scale fusion architecture, featuring enhancing low light image with output calibration controls.



MOTIVATION



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Research based
Image restoration

Evaluating model
generalization



FEATURES



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User
adjustable
preference

Users can input
multiple images
at a time

Download
option

Simple UI



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02.

METHODOLOGY



Features of the project



COMPONENTS

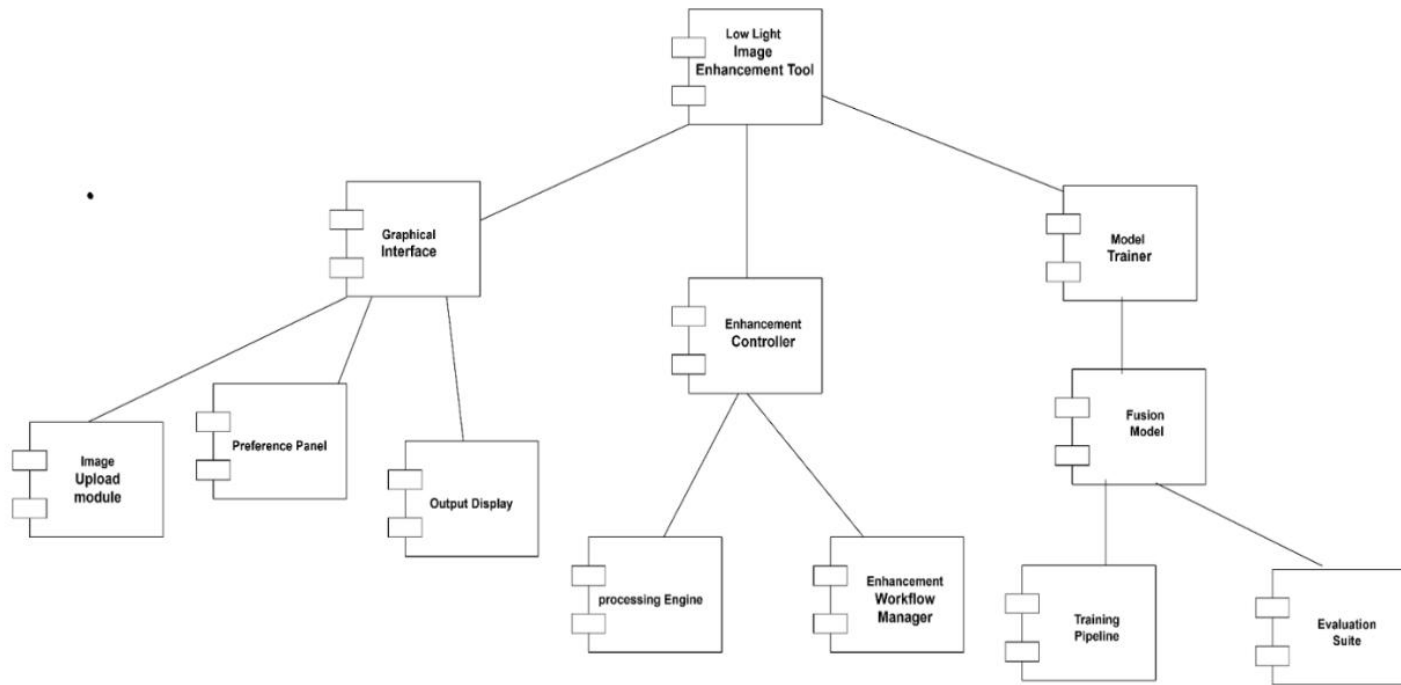


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Model architecture

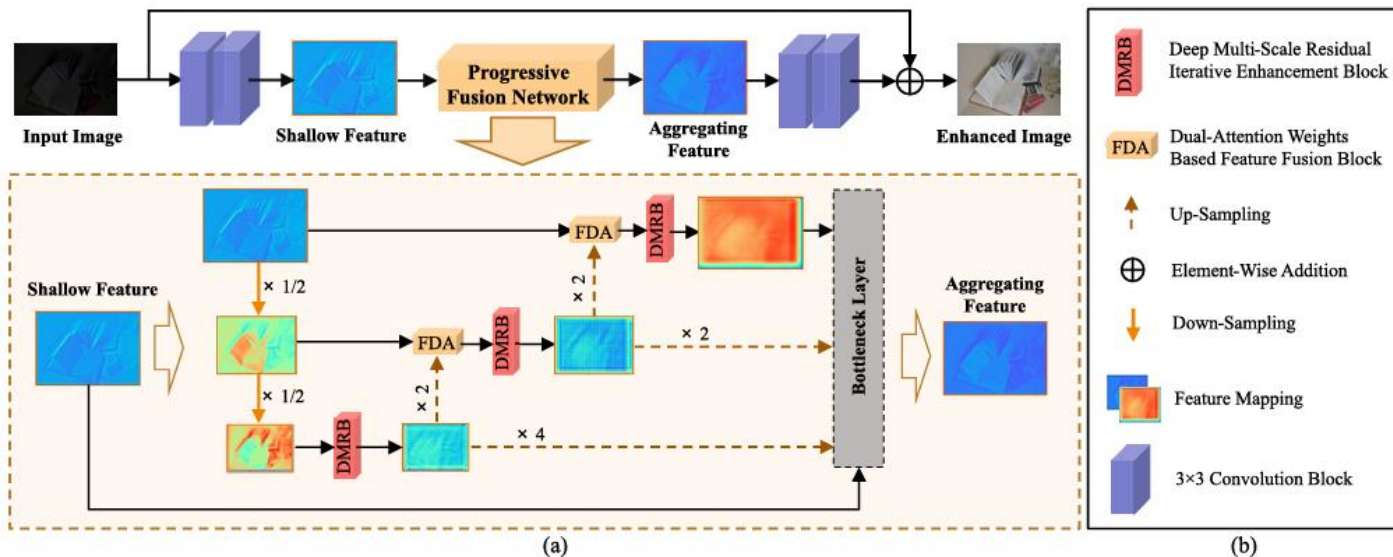


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SYSTEM OVERVIEW



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$$\hat{I}_{final} = \mathcal{O}(\mathcal{P}(\mathcal{I}(I_{raw})))$$

Where:

- $\mathcal{I}$ = input preprocessing
- $\mathcal{P}$ = DRSformer processing
- $\mathcal{O}$ = output refinement



BRAIN OF THE PROJECT



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Self-attention formulation:

$$\mathcal{A}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V$$

Sparse attention selection introduces masking:

$$\mathcal{A}_{\text{sparse}}(Q, K, V) = \text{Softmax} \left(\mathcal{M} \odot \frac{QK^T}{\sqrt{d}} \right) V$$

Where mask $\mathcal{M}$ is defined as:

$$\mathcal{M}_{ij} = \begin{cases} 1, & \text{if } (QK^T)_{ij} \in \text{Top-}k \\ 0, & \text{otherwise} \end{cases}$$



Tuning



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Gamma refinement:

$$\hat{I}_{\gamma} = \hat{I}^{\gamma}$$

Contrast adjustment:

$$\hat{I}_{final} = \alpha \hat{I}_{\gamma} + \beta$$

Where:

- γ : gamma correction factor
- α : contrast parameter
- β : brightness offset

The model is trained by minimizing a composite loss function:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{char} + \lambda_2 \mathcal{L}_{ssim} + \lambda_3 \mathcal{L}_{color} + \lambda_4 \mathcal{L}_{tv}$$



Dataset



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- Dataset: LOL (Low-Light) dataset (v1 and v2).
- Structure: Paired images consisting of low-light inputs and normal-light ground truths.
- Split: 90% for training and 10% for validation/testing.
- Purpose: Providing a supervised learning environment to teach the model lighting restoration.



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03.

RESULTS



How much has it improved?



Outcome



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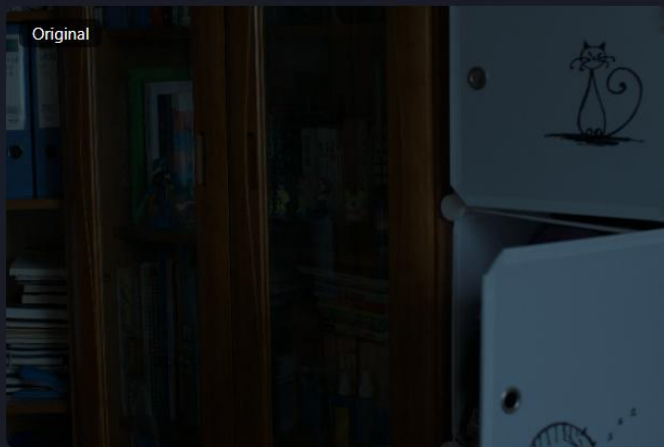
04



1.png

DONE ✓

Original



Enhanced



600 × 400 px

Download



COMPARISON

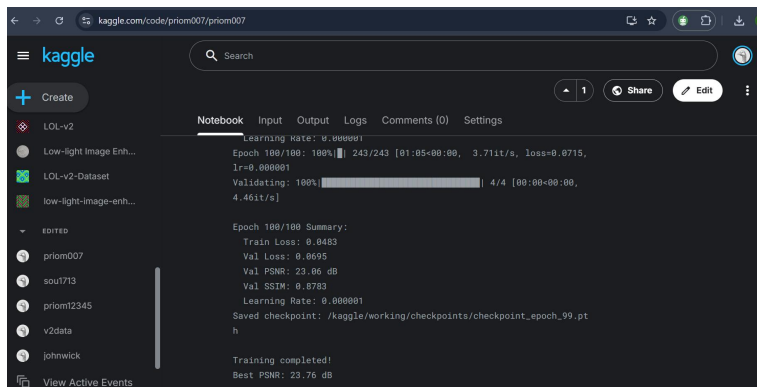


01

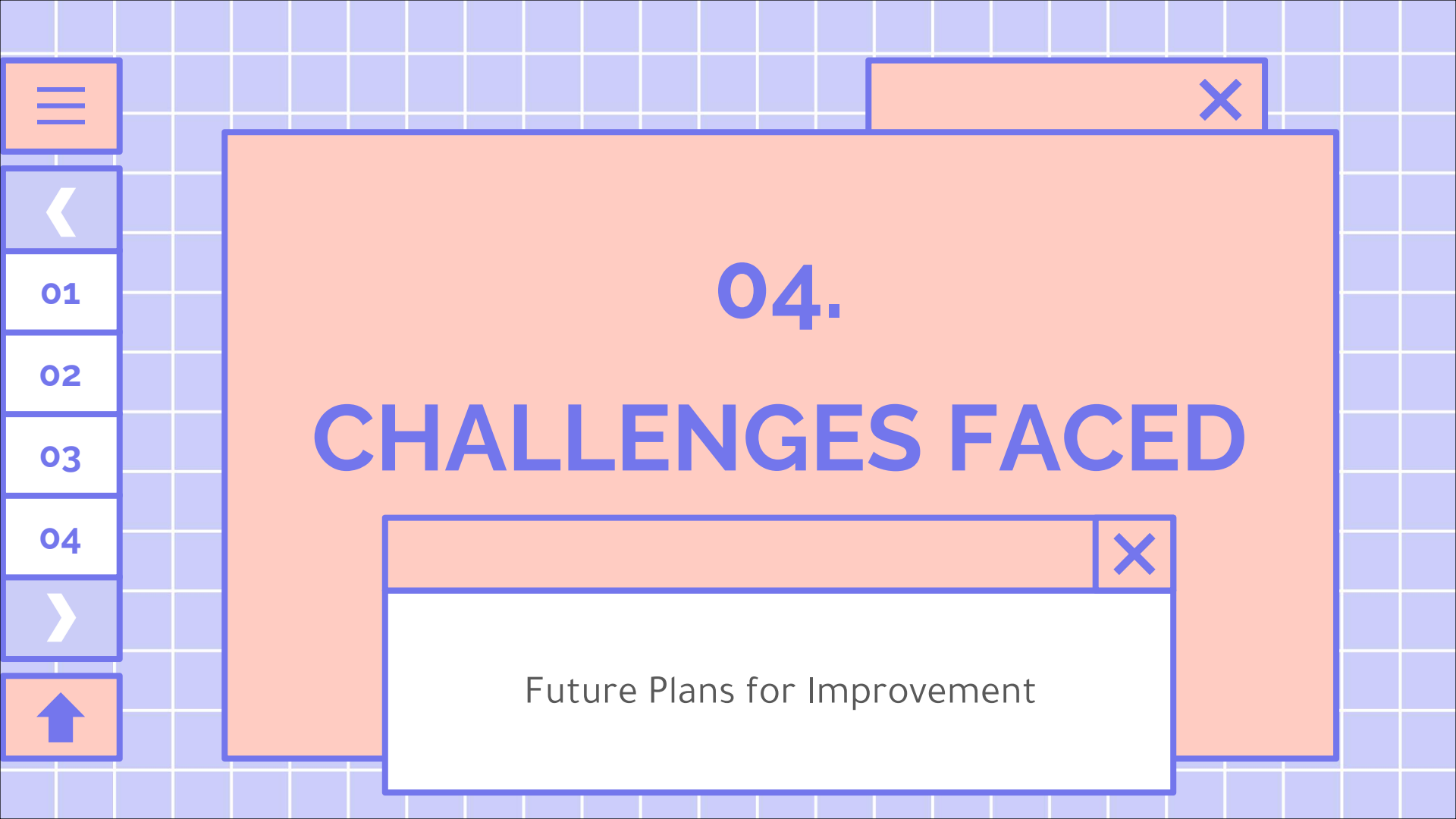
02

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04



Metrics	LIME	RetinexNet	EnlightenGAN	Zero-DCE	DLN	RUAS	SCI	ALL-E	PairLIE	Ours
PSNR↑	15.9653	19.3847	19.6945	16.7554	21.392	18.4507	16.4467	20.2486	<u>21.7592</u>	22.5665
SSIM↑	0.5769	0.5594	0.6578	0.5848	0.8095	0.6668	0.5229	<u>0.8159</u>	0.8102	0.8256
LPIPS↓	0.372	0.474	0.322	0.335	0.300	0.270	0.340	0.230	<u>0.248</u>	0.287
NIQE↓	8.0912	8.8787	<u>4.6839</u>	7.7669	6.0917	6.3402	7.8766	7.2295	4.0851	5.9036

The slide features a light blue grid background. On the left is a vertical sidebar with six buttons: a menu icon (three horizontal lines), a back arrow, a list of numbers 01 through 04 (with 04 highlighted), a forward arrow, and an up arrow. The main content area is a large light orange rectangle with a blue border, containing the title '04. CHALLENGES FACED' in blue text. At the top right of this area is a small orange box with a blue 'X'. At the bottom is a white box with a blue border containing the text 'Future Plans for Improvement'.

04. CHALLENGES FACED

Future Plans for Improvement



CHALLENGES FACED



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Complexity
of Hybrid
Architecture

Training

Hardware
Limitation

Output
Calibration

Batch
processing

Researching



Future Scope & Limitations



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Scope

1. Unsupervised learning
2. Real-time video

Limitations

1. Hardware dependency
2. Resolution Constraints
3. Dataset bias
4. Static image only

References

[1] Zhang, Hongxin, Teng Ran, Wendong Xiac, Kai Lv, Song Peng, Liang Yuan, and Jingchuan Wang. "Multi-Scale Progressive Fusion Network for Low-Light Image Enhancement." IEEE Transactions on Instrumentation and Measurement (2025).

[2]Zhang, Kaibing, Cheng Yuan, Jie Li, Xinbo Gao, and Minqi Li. "Multi-branch and progressive network for low-light image enhancement." IEEE Transactions on Image Processing 32 (2023): 2295-2308.

[3]CHATGPT(for image of formula)



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THANKS!

Do you have any questions?

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